Find Original Array From Doubled Array (medium)

Problem Statement

A doubled array `changed` is formed from the `original` array after appending twice the value of each element in the `original` array and `randomly shuffling` elements in the updated `original` array.

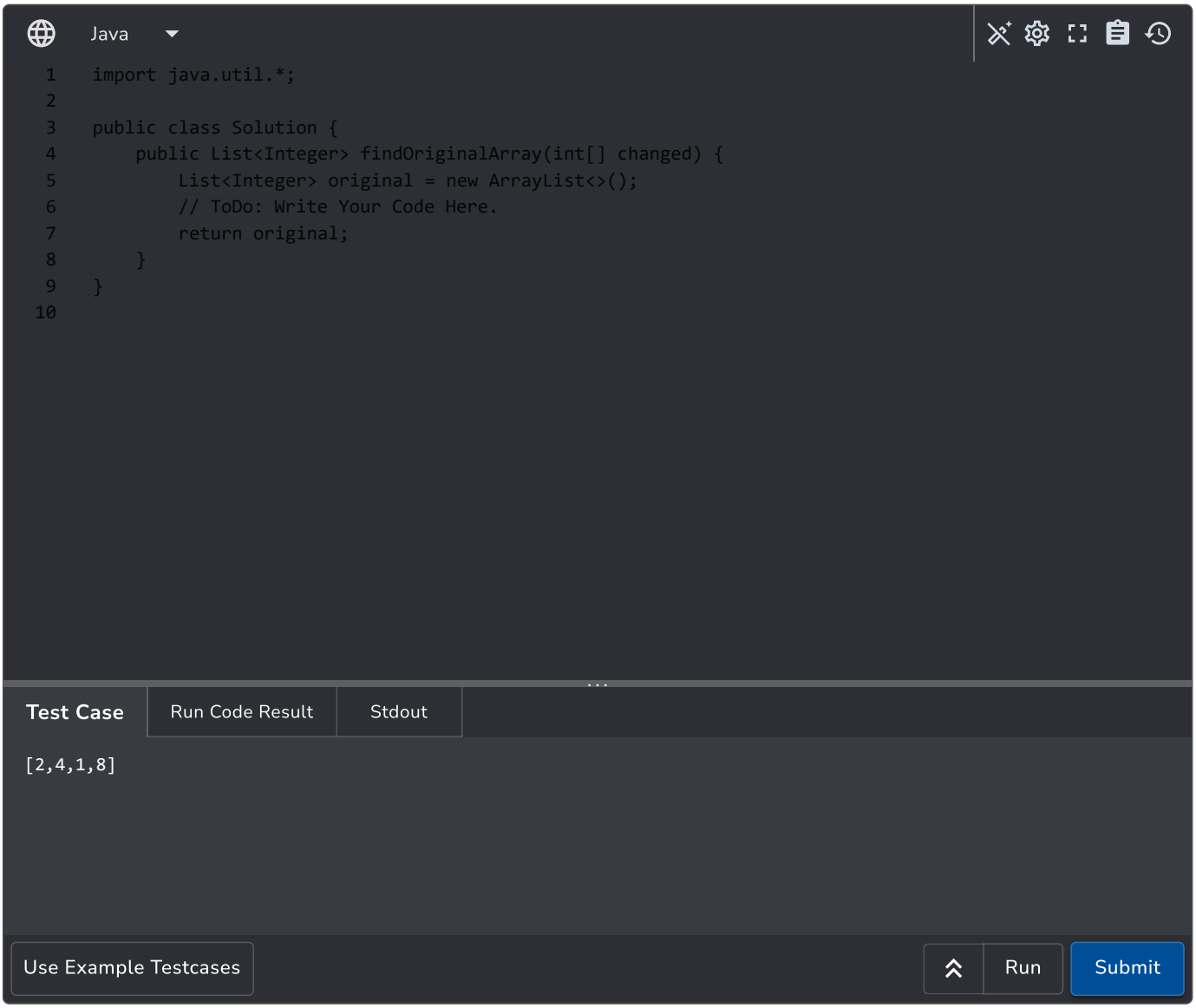
Given an array `changed`, return the `original` array if `changed` is a valid `doubled` array. Otherwise, return `empty` array.

Examples

- **Example 1**:
  - **Input**: `changed = [2,4,1,8]`
  - **Expected Output**: `[1,4]`
  - **Justification**: The original array [**1,4**] is doubled to form [**2,4,1,8**] (1's double is 2, and 4's double is 8).
- **Example 2**:
  - **Input**: `changed = [6,3,0,0]`
  - **Expected Output**: `[0,3]`
  - **Justification**: The original array [**0,3**] is doubled to form [**6,3,0,0**] (0's double is 0, and 3's double is 6).
- **Example 3**:
  - **Input**: `changed = [1, 2, 2, 4, 4, 8, 16, 32]`
  - **Expected Output**: `[1, 2, 4, 16]`
  - **Justification**: The original array [**1, 2, 4, 16**] is doubled to form [**1, 2, 2, 4, 4, 8, 16, 32**] (1's double is 2, 2's double is 4, 4's double is 8, and 16's double is 32).

Try it yourself

Try solving this question here:



← Previous

Solution: Restore IP Addresses

Next →

Solution: Find Original Array From Doubled Array

Mark as Completed